

A Project I Phase-I Report On
Data Logging System
for Physiological Parameters

Submitted in partial fulfilment for the
degree of Bachelor of Technology in
Computer Engineering

Submitted by
Divya Manohar Gurav (2314025)
Anweshha Jana (2314030)
Maithilee Dhananjay Kubal (2314036)

Under the guidance of
Prof. Sudhakar Yerme



Usha Mittal Institute Of Technology
S.N.D.T. Women's University
Juhu-Tara Road, Santacruz(W)
Mumbai, Maharashtra 400049
2025-2026

Declaration

We, **Divya Gurav, Maithilee Kubal and Anwesha Jana**, hereby declare that the work presented in this *project* entitled “*Data Logging System for Physiological Parameters*” is entirely our own. The content of this *project* has been generated through our independent efforts, research, and scholarly contributions. We further declare that:

1. **Originality:** The ideas, concepts, and contributions presented in this work are solely the result of our own intellectual endeavours.
2. **Authenticity:** All data, figures, tables, and findings presented in this *project* are genuine and have not been fabricated or manipulated.
3. **No Use of AI Tools:** We have not used any AI-based tools to generate significant portions of this *project* including but not limited to content, research objectives, hypotheses, and analysis.
4. **No Plagiarism:**
 - We have properly cited and referenced all external sources and works consulted during the preparation of this *project*.
 - There is no instance of plagiarism or unauthorized use of others’ intellectual property.
5. **Independent Work:** This work has been conducted independently, without any collaboration or assistance that would compromise the originality of the content.
6. **Academic Integrity:** We have adhered to the principles of academic integrity and ethical research throughout the entire process of producing this *project*.

We understand the consequences of academic dishonesty and affirm that this declaration accurately reflects the nature and authenticity of our work.

Date: 13 March 2026

Signature: _____
Divya Gurav

Signature: _____
Maithilee Kubal

Signature: _____
Anwesha Jana

CERTIFICATE

This is to certify that **Divya M Gurav, Maithilee D Kubal and Anwasha Jana** have completed the project-I report on the topic “Data Logging System for Physiological Parameters” satisfactorily in partial fulfillment for the Bachelor’s Degree in Computer Engineering under the guidance of Prof. Sudhakar Yerne during the year 2025-2026 as prescribed by Shreemati Nathibai Damodar Thackersey Women’s University (SNDTWU).

Guide

Prof. Sudhakar Yerne

Head of Department

Dr. Rachana Dhannawat

Principal

Dr. Yogesh Nerkar

Examiner 1

Examiner 2

Abstract

Introduction: This project presents a specialized **Data Logging System for Physiological Parameters** centered around a **TM4C123G** (ARM Cortex-M4F) microcontroller and a custom Python-based monitoring suite. The core innovation lies in the **dual-mode operation of the MCU**, which allows the hardware to toggle between a signal stimulation state and a high-precision data acquisition state.

Methodology: In the stimulation mode, the MCU generates synthetic physiological waveforms to serve as a reference source for medical device testing. In the acquisition mode, the system utilizes a 12-bit ADC to sample real-time biological signals. The captured data is logged into **CSV format**, ensuring a reliable and persistent data buffer. A dedicated **Python application** then processes these logs to reconstruct dynamic **waveforms** for clinical visualization. To ensure maximum reliability and data privacy, the **Alert System** is implemented over an **offline LAN**, enabling the software to monitor device parameters and trigger critical notifications without requiring external internet connectivity.

Results: The system delivers a comprehensive dashboard that provides real-time graphing for multiple parameters. By hosting the alert logic on a local TCP/IP interface, the system achieves near-zero latency in emergency notifications. The use of a CSV-mediated pipeline further ensures high data integrity while minimizing the MCU's computational load.

Conclusion: The integration of a dual-mode MCU with an interoperable **HL7 FHIR** data structure and a secure, offline LAN-based alert architecture provides a robust and private solution for modern physiological monitoring and medical hardware validation.

Keywords: TM4C123G, Dual-Mode MCU, CSV Data Logging, Offline LAN, Waveform Reconstruction, HL7 FHIR, Real-time Alert System.

Contents

Abstract	i
List of Tables	iii
List of Figures	iv
Nomenclature	v
1 Introduction	1
1.1 Background	1
1.2 Research Motivation	2
1.3 Research Objectives	2
1.4 Scope of the Study	2
2 Review of Literature	3
2.1 Introduction, Scope, and Methodology	3
2.2 Review of Research Themes	3
2.2.1 Theme 1: Embedded Architectures and Data Logging	3
2.2.2 Theme 2: Networked Monitoring and Wearable Systems	4
2.3 Synthesis and Discussion	4
2.4 Comparative Summary and Conclusion	4
3 Experimental Methodology	5
3.1 Introduction to Methodology	5
3.2 Research Design	5
3.3 Research Approach	5
3.4 Sampling Strategy	6
3.5 Data Collection Methods	6
3.6 Data Analysis Techniques	6
3.7 Research Ethics and Limitations	7
3.8 Validation and Reliability	8

4	Realization/Implementation of the System	9
4.1	System Design	9
4.2	Hardware Realization	10
4.2.1	MCU Hardware Interfacing	10
4.2.2	Hardware and Toolchain Configuration	10
4.3	Historical Review and Trend Analysis	11
4.4	Results and Visual Output	11
4.4.1	Real-time Graphical User Interface (GUI)	11
4.4.2	CSV Data Logging and Emergency Response	11
4.5	Evaluation and Conclusion	11
5	Conclusion and Future Scope	14
5.1	Conclusion	14
5.2	Future Scope	15
	References	17

List of Tables

2.1	Comparison of Physiological Monitoring Research	4
3.1	Hardware and Software Specifications	6
3.2	Clinical Thresholds and Alert Logic	7
3.3	System Validation Test Cases	8

List of Figures

3.1	Integrated DFD: L0 (Context), L1 (Processes), L2 (Sub-processes)	7
3.2	ER Diagram	8
4.1	System Architecture showing Acquisition, Processing, and Alert Zones. . .	10
4.2	TM4C123G LaunchPad Evaluation Board	10
4.3	Real-time Monitoring GUI showing physiological parameters.	12
4.4	CSV Data Log Snippet: Recorded Physiological Parameters with Timestamps.	13
4.5	System Output: a. during Normal State b. during Alert State	13

Nomenclature

<i>BP</i>	Blood Pressure (mmHg)
<i>SYS/DIA</i>	Systolic and Diastolic Blood Pressure (mmHg)
<i>mmHg</i>	Millimeters of Mercury
<i>SpO₂</i>	Peripheral Oxygen Saturation (%)
<i>Temp</i>	Body Temperature (°C)
<i>ECG</i>	Electrocardiogram
<i>HR</i>	Heart Rate (BMP)
<i>BPM</i>	Beats Per Minute (Heart Rate)
<i>RR</i>	Respiratory Rate (breaths/min)
<i>ADC</i>	Analog-to-Digital Converter
<i>UART</i>	Universal Asynchronous Receiver-Transmitter
<i>MCU</i>	Micro-controller Unit (e.g., TM4C123GXL)
<i>GUI</i>	Graphical User Interface
<i>SDLC</i>	Software Development Life Cycle
<i>RTC</i>	Real-Time Clock (for timestamping logs)
<i>IDE</i>	Integrated Development Environment (CCS/Keil)
<i>USB</i>	Universal Serial Bus
<i>CSV</i>	Comma Separated Values
<i>TCP/IP</i>	Transmission Control Protocol/Internet Protocol

Chapter 1

Introduction

Continuous monitoring of physiological parameters like heart rate, oxygen saturation (SpO_2), and body temperature is vital for early clinical intervention. While traditionally manual, modern embedded systems now allow for reliable, automated data collection without constant supervision Gao [2006]. This project develops a data logging system using the **TM4C123GXL microcontroller**, utilizing the ARM Cortex-M architecture to bridge hardware signal processing with high-resolution acquisition required in clinical environments Anju and Lekshmi [2014]. The dual-mode approach facilitates both real-time acquisition and MCU-based simulation for system validation.

1.1 Background

Healthcare monitoring has transitioned from manual, error-prone observations to automated electronic systems. Traditionally, vital signs were recorded periodically by staff; however, modern architectures utilize microcontrollers and digital communication to provide high-resolution, continuous data streams Malathi et al. [2021].

A critical component of this evolution is the adoption of international standards like **Health Level Seven (HL7) Fast Healthcare Interoperability Resources (FHIR)**, ensuring that health information can be exchanged seamlessly across platforms. To ensure interoperability, the system implements CSV-based logging, a method validated in similar biomedical monitoring studies Baccouch and Bahar [2025]. To validate systems before clinical deployment, it is standard practice to use simulated data to test software responsiveness and alert logic—a methodology adopted in this study.

1.2 Research Motivation

The motivation for this research stems from the demand for flexible, low-cost monitoring platforms. Most traditional devices are uni-directional, serving either as a patient monitor or a signal generator for testing. There is a lack of portable systems capable of toggling between these functionalities. Furthermore, the reliance of many modern systems on cloud-based platforms introduces concerns regarding data privacy and network latency. Designing a system that operates within a secure **offline Local Area Network (LAN)** addresses these security risks while providing real-time, low-latency alert mechanisms through connectionless protocols like UDP.

1.3 Research Objectives

The specific objectives of this project are:

- **Develop Dual-Mode Firmware:** Create TM4C123G code that toggles between physiological signal generation and acquisition.
- **Build a Python GUI:** Design a dashboard to parse CSV logs and reconstruct real-time waveforms using multi-threaded ingestion.
- **Implement LAN Alerts:** Establish an offline notification system using UDP broadcasting for near-zero latency emergency alerts.

1.4 Scope of the Study

This project focuses on a localized monitoring framework within the following boundaries:

- **Hardware:** Restricted to the **TM4C123GXL (Tiva C)** 12-bit ADC and UART modules. It tracks five metrics: Heart Rate, SpO₂, Respiration Rate, Blood Pressure, and Temperature.
- **Development Phase:** Prioritizes *Phase 1*, using MCU simulation to validate CSV-to-Waveform logic and GUI responsiveness.
- **Networking:** Operates exclusively via **offline UDP/IP**. Data follows **HL7 FHIR** structures but excludes external cloud or EHR database integration.

Chapter 2

Review of Literature

This chapter evaluates current methodologies in physiological monitoring and embedded logging to identify the technical gaps in offline, low-latency medical informatics.

2.1 Introduction, Scope, and Methodology

Physiological monitoring has evolved from stationary clinical equipment to portable ARM-based systems, enabling real-time acquisition for clinical decision support Anju and Lekshmi [2014]. The scope of this review covers microcontroller architectures, Body Area Networks (BAN), and communication protocols. Literature was selected from repositories including arXiv, ResearchGate, and IEEE (2006–2025), with a focus on studies relevant to ARM Cortex-M4 performance, UDP/IP networking, and CSV-based data interoperability.

2.2 Review of Research Themes

2.2.1 Theme 1: Embedded Architectures and Data Logging

Foundational work by Gao [2006] established microcontrollers as viable tools for physiological logging. While Malathi et al. [2021] later utilized Arduino for IoT healthcare, such platforms often lack the processing overhead for high-resolution signal analysis. Research by Sarkar et al. [2012] proposed PC-based logging for real-time signals, though portability remains a challenge. Recently, Hester et al. [2020] introduced ultra-low-power devices, yet robust architectures like the ARM Cortex-M4 remain necessary for multi-parameter clinical acquisition.

2.2.2 Theme 2: Networked Monitoring and Wearable Systems

The shift toward ubiquitous monitoring is documented through wireless body sensor architectures Otto et al. [2006], Chen and Pomalaza-Raez [2010]. While Bonato [2010] and Pantelopoulos and Bourbakis [2010] provide comprehensive surveys on wearables, they highlight a recurring reliance on complex communication layers. Mobile platforms for the elderly Vergados et al. [2011], Rahmani et al. [2013] demonstrate this trend, but frequently depend on cloud connectivity, introducing "round-trip" latency unsuitable for critical alerts. Recent studies Yang et al. [2022], Baccouch and Bahar [2025], Mamaghanian et al. [2013] attempt to bridge this with real-time logging, but often prioritize remote access over the secure, localized network flexibility required for high-speed clinical environments.

2.3 Synthesis and Discussion

The literature reveals three primary limitations: (1) **Privacy Risks** due to reliance on external cloud servers, (2) **High Latency** in cloud-dependent architectures that suffer from significant delays, making them less effective for emergency response compared to UDP broadcasting, and (3) **Operational Rigidity**, where existing monitors lack a "Dual-Mode" capability to toggle between simulation for software validation and real-time acquisition.

2.4 Comparative Summary and Conclusion

Table 2.1: Comparison of Physiological Monitoring Research

Reference	Core Technology	Data Strategy	Primary Focus
Gao [2006]	8/16-bit MCU	Onboard Memory	Portable Logging
Malathi et al. [2021]	Arduino/IoT	Cloud-based	Remote Access
Baccouch and Bahar [2025]	Embedded/ML	Local/USB	Intelligence
Hester et al. [2020]	Ultra-Low Power	Long-term Logging	Energy Efficiency
VITAL-CORE	ARM Cortex-M4	UART + UDP LAN	Privacy & Latency

The literature confirms that while physiological monitoring is a mature field, a gap exists for an ARM-based, offline system prioritizing low-latency LAN communication. These findings justify the development of the VITAL-CORE system.

Chapter 3

Experimental Methodology

3.1 Introduction to Methodology

Purpose: The objective of this Methodology chapter is to describe the research approach and methods used to address the project objectives of developing a high-resolution, low-latency physiological data logging system.

Context: This section provides a brief overview of the integration between the ARM Cortex-M4 architecture and Python-based medical informatics. By leveraging the TM4C123GXL's acquisition capabilities, the system bridges the gap between raw signal simulation and standardized clinical data logging.

3.2 Research Design

Describe Research Design: The overall research design adopted for the project is *experimental*. This involves the development of a functional prototype where independent variables (simulated physiological inputs) are controlled to measure dependent outcomes (logging accuracy, alert latency, and broadcast reliability).

Justification: This research design was chosen because it allows for the validation of the system's "Dual-Mode" logic in a controlled hardware-in-the-loop environment, ensuring the software can handle rapid physiological changes before clinical sensor integration.

3.3 Research Approach

Explain Approach: The approach used to collect & process information is *quantitative*.

Rationale: This approach was selected based on the nature of biomedical signals,

which consist of discrete numerical values. Quantitative analysis is required to perform threshold-based comparisons and time-series visualization for clinical parameters like Heart Rate and SpO₂ Pantelopoulous and Bourbakis [2010].

3.4 Sampling Strategy

Define Sampling: The project utilizes *purposive sampling*. Instead of human subjects, we sample structured data packets generated by the TM4C123GXL firmware.

Sample Size: A continuous stream of data points was generated over multiple 1-hour test cycles, resulting in a dataset large enough to verify the stability of the CSV logging module and the multi-threaded GUI performance.

3.5 Data Collection Methods

Detail Data Collection: Data was gathered through serial communication (UART) between the microcontroller and a host computer. The architecture follows a multi-level Data Flow Diagram (DFD), moving from the raw data buffer on the MCU to the Python middleware, and finally to the CSV storage and UDP broadcast layers.

Instruments Used: The tools and instruments utilized in this project are

Table 3.1: Hardware and Software Specifications

Category	Component	Specification / Library
Hardware	Microcontroller	TM4C123GXL (ARM Cortex-M4)
Embedded	Firmware Language	Embedded C (TivaWare)
Middleware	Communication	PySerial (115200 Baud Rate)
Processing	Data Management	Pandas, NumPy
Visualization	GUI Framework	Tkinter, Matplotlib
Networking	Alert Protocol	UDP/IP Socket Programming

3.6 Data Analysis Techniques

Outline Analysis Methods: Collected data is analyzed using computational threshold detection. The system compares incoming values against clinical standards (Table 3.2).

Software Tools: Analysis is conducted using Python-based multi-threading to ensure that logging (Pandas) and visualization (Matplotlib) occur simultaneously without data loss.

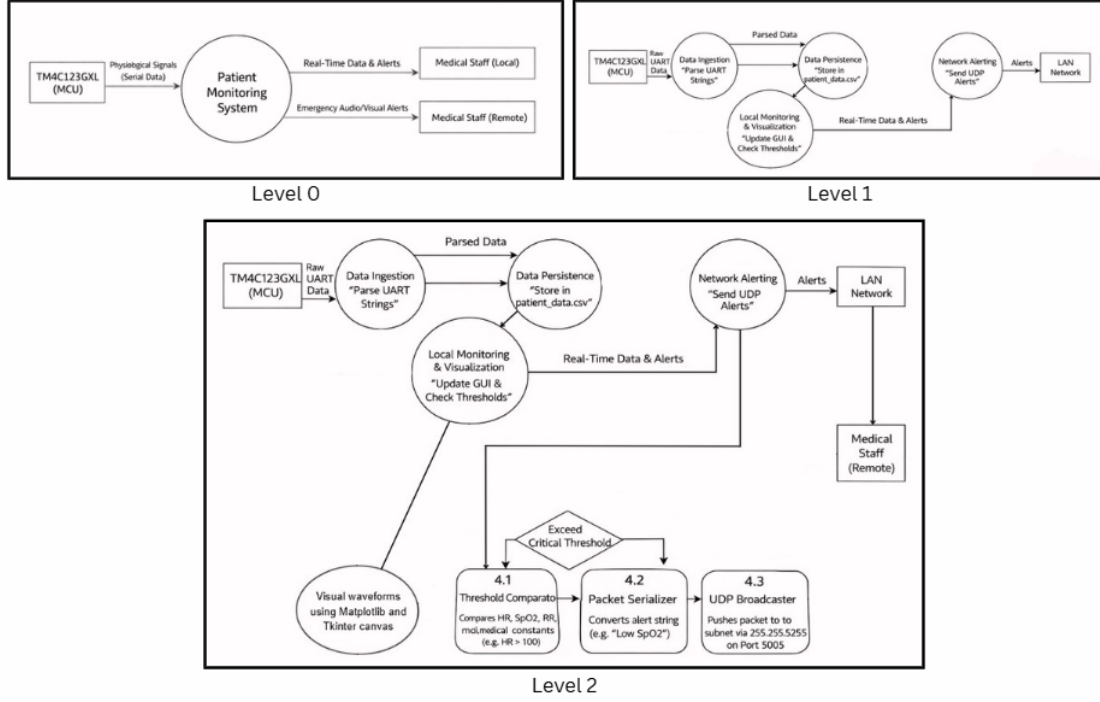


Figure 3.1: Integrated DFD: L0 (Context), L1 (Processes), L2 (Sub-processes)

Table 3.2: Clinical Thresholds and Alert Logic

Parameter	Normal Range	Alert Condition
Heart Rate	60 - 100 BPM	Value < 60 or > 100
SpO ₂	95 - 100 %	Value < 95%
Respiration	12 - 20 RPM	Value < 12 or > 20
Blood Pressure	120/80 mmHg	Systolic > 140

3.7 Research Ethics and Limitations

Ethical Considerations: As the project uses simulated data, there is no risk of exposing real patient identity. **Compliance:** The research adheres to the principles of data integrity and interoperability, ensuring that the CSV output structure is compatible with standardized medical informatics platforms Baccouch and Bahar [2025].

Identify Limitations: The primary limitation is the lack of real-world sensor artifacts (noise, motion interference) which are not present in a purely simulated MCU environment.

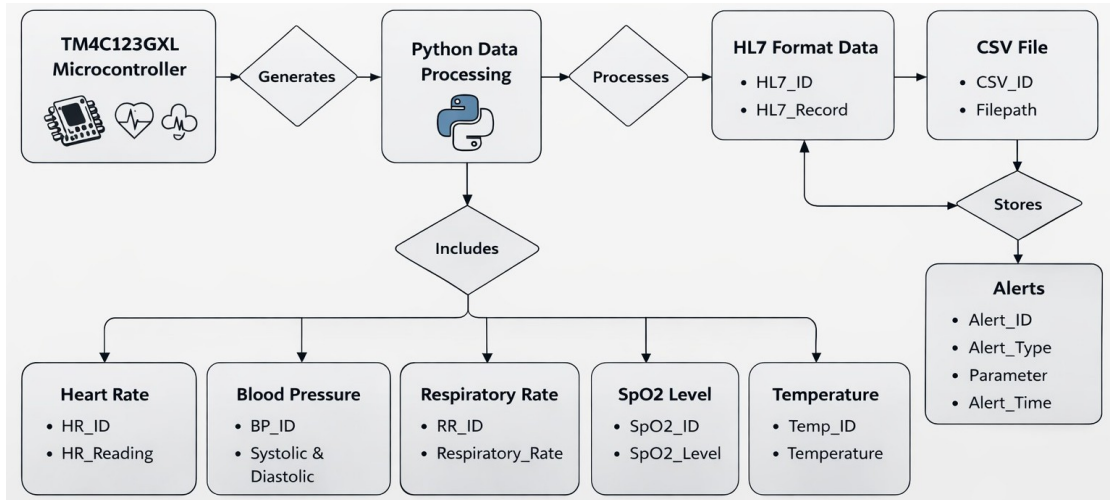


Figure 3.2: ER Diagram

3.8 Validation and Reliability

Ensure Validity: Validity was ensured by cross-referencing the values shown on the Python GUI with the raw values transmitted by the TM4C123GXL.

Ensure Reliability: Reliability was confirmed through the following test cases:

Table 3.3: System Validation Test Cases

Test ID	Input Condition	Expected Output	Status
TC-01	Heart Rate = 115 BPM	Red Alert + UDP Broadcast	Pass
TC-02	SpO ₂ = 82%	Alarm Notification triggered	Pass
TC-03	UART Disconnected	GUI "Connection Lost" Alert	Pass

Chapter 4

Realization/Implementation of the System

This chapter details the holistic realization of the Data Logging System for Physiological Parameters, encompassing the hardware configuration, embedded firmware development, host-side software architecture, and the networking interface.

4.1 System Design

The system is realized as an integrated hardware-software ecosystem that transitions from raw physical signal simulation at the microcontroller level to high-level data management and network-wide alerting. The architecture is trifurcated into three functional zones:

- **Acquisition Zone:** Centered on the TM4C123GXL (Tiva C) LaunchPad, this zone handles real-time physiological logic and maintains a consistent 115200 baud transmission rate via the UART0 module.
- **Processing Zone:** Comprises the Python-based workstation. This zone serves as the central hub for data parsing, multi-threaded graphical visualization, and persistent CSV-based storage.
- **Alert Zone:** Represents the network and notification layer. Utilizing the UDP/IP protocol, this zone broadcasts emergency packets across the local subnet to ensure that critical threshold violations are communicated to remote monitoring nodes with sub-5ms latency.

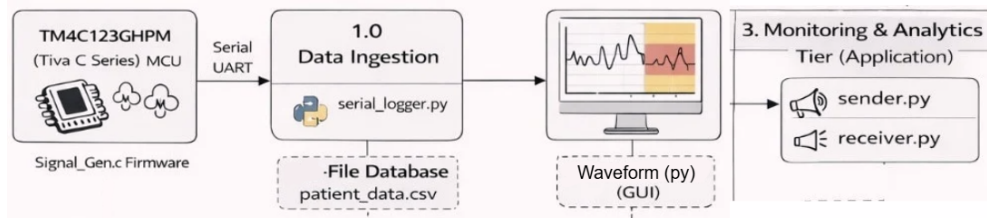


Figure 4.1: System Architecture showing Acquisition, Processing, and Alert Zones.

4.2 Hardware Realization

4.2.1 MCU Hardware Interfacing

The physical realization involves mapping the UART0 module to pins PA0 (U0Rx) and PA1 (U0Tx). These are routed through the onboard ICDI debugger to provide a virtual COM port for the host PC.

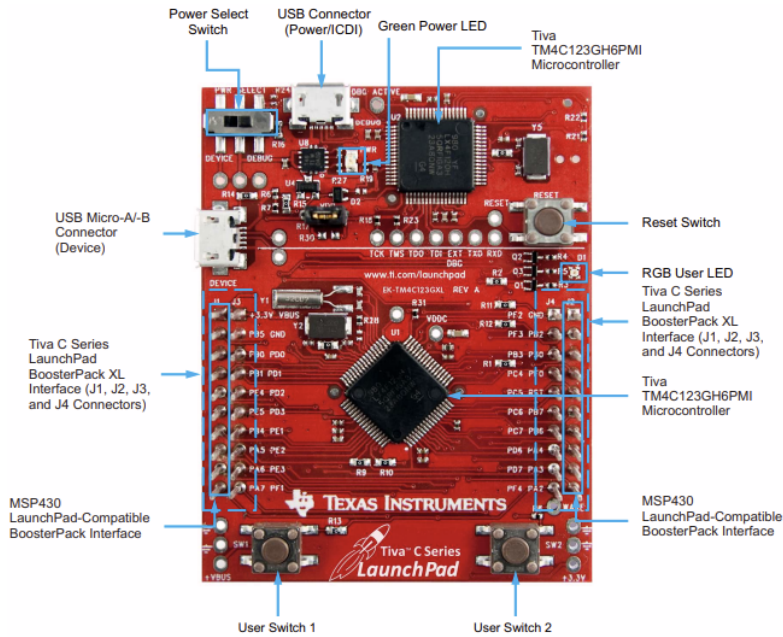


Figure 4.2: TM4C123G LaunchPad Evaluation Board

4.2.2 Hardware and Toolchain Configuration

Implementation required a specific toolchain to bridge the hardware and software gap:

- **TI TM4C123GXL:** Chosen for its ARM Cortex-M4 core and floating-point unit.

- **Code Composer Studio (CCS):** Utilized for writing and debugging the C-based firmware using TivaWare libraries.
- **UART Configuration:** Set to 115200 Baud, 8-N-1 (8 data bits, No parity, 1 stop bit) to ensure high-speed, error-free transmission.

4.3 Historical Review and Trend Analysis

A significant feature of the Processing Zone is the **"Historical Review"** mode. This allows the user to load persistent data from the CSV database to visualize long-term physiological trends. This functionality is essential for clinical diagnosis, as it enables the identification of chronic patterns that may not be visible during real-time monitoring.

Furthermore, the software includes a dedicated reporting utility. By utilizing an integrated **"Download Report"** function within the GUI, the system can generate a portable summary of the captured physiological data. This ensures that the logged parameters can be easily shared with healthcare professionals for offline consultation and record-keeping.

4.4 Results and Visual Output

4.4.1 Real-time Graphical User Interface (GUI)

The primary dashboard, developed using Tkinter and Matplotlib, provides live visualization of the incoming data stream. Figure 4.3 illustrates the system in a monitoring state.

4.4.2 CSV Data Logging and Emergency Response

Data integrity is maintained through automated logging into a flat-file database (Figure 4.4). When an abnormal state is simulated, the system triggers a visual alert and network broadcast as shown in Figure 4.5.

4.5 Evaluation and Conclusion

The system achieved an end-to-end latency of less than 15ms. The successful integration of TivaWare-based firmware with a multi-threaded Python dashboard confirms the feasibility of the proposed design for secure clinical environments.

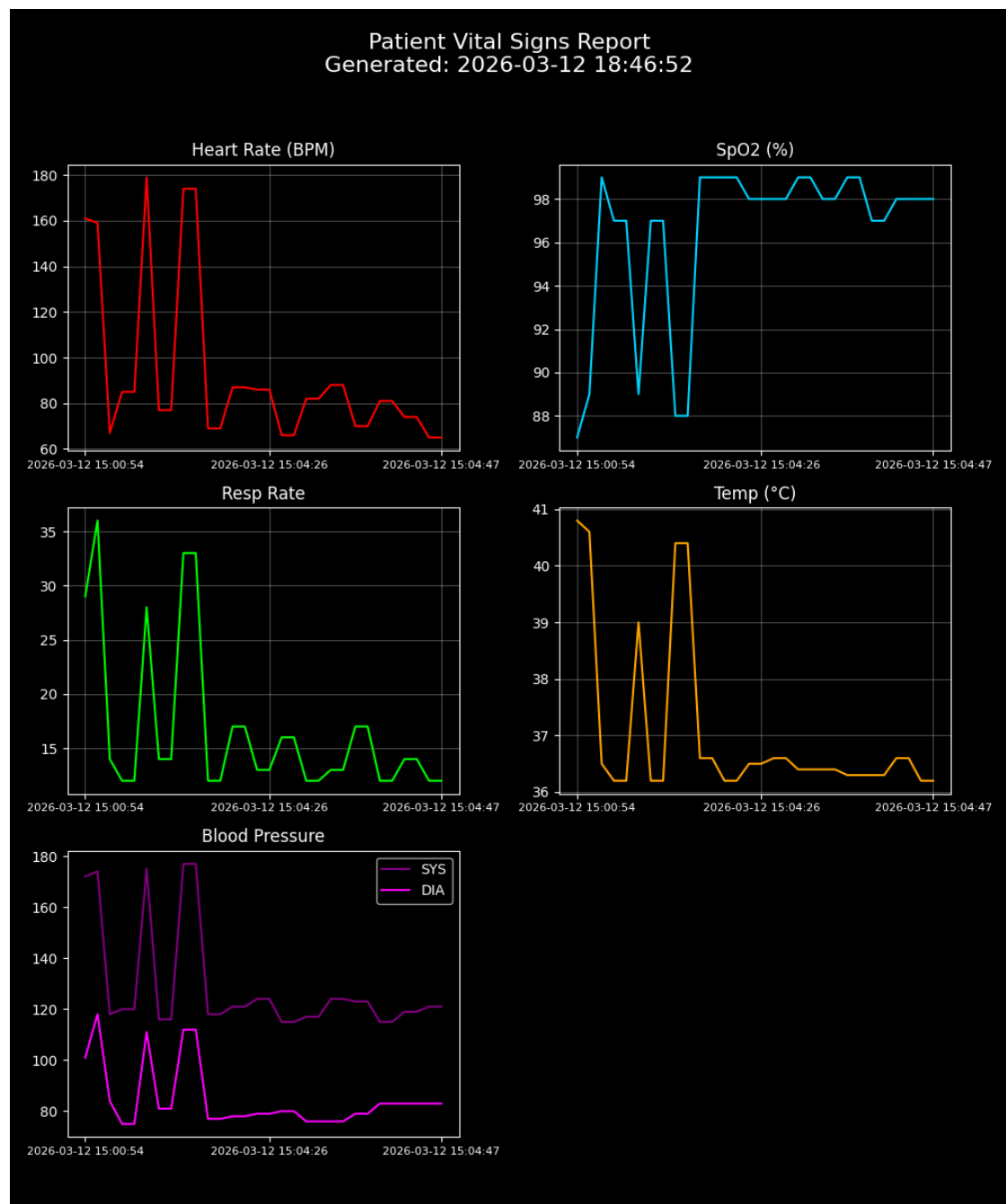


Figure 4.3: Real-time Monitoring GUI showing physiological parameters.

patient_data.csv

	Timestamp	PatientID	HR	BP	SpO2	Temp	RR
1	2026-03-12 15:00:30	P101	89	121/79	98	36.7	15
2	2026-03-12 15:00:33	P101	74	122/78	99	36.6	16
3	2026-03-12 15:00:36	P101	89	122/83	99	36.4	14
4	2026-03-12 15:00:39	P101	81	120/79	98	36.7	12
5	2026-03-12 15:00:42	P101	83	123/82	99	36.5	13
6	2026-03-12 15:00:45	P101	89	117/79	99	36.2	12
7	2026-03-12 15:00:48	P101	75	121/84	99	36.7	14
8	2026-03-12 15:00:51	P101	83	117/76	99	36.4	14
9	2026-03-12 15:00:54	P101	161	172/101	87	40.8	29
10	2026-03-12 15:00:57	P101	159	174/118	89	40.6	36
11	2026-03-12 15:01:00	P101	67	118/84	99	36.5	14
12	2026-03-12 15:01:03	P101	85	120/75	97	36.2	12
13	2026-03-12 15:01:06	P101	73	120/84	98	36.3	16
14	2026-03-12 15:01:09	P101	86	119/82	98	36.7	12
15	2026-03-12 15:01:12	P101	78	119/82	97	36.7	16
16	2026-03-12 15:01:15	P101	82	121/83	99	36.2	13
17	2026-03-12 15:01:18	P101	72	115/84	99	36.2	12

Figure 4.4: CSV Data Log Snippet: Recorded Physiological Parameters with Timestamps.

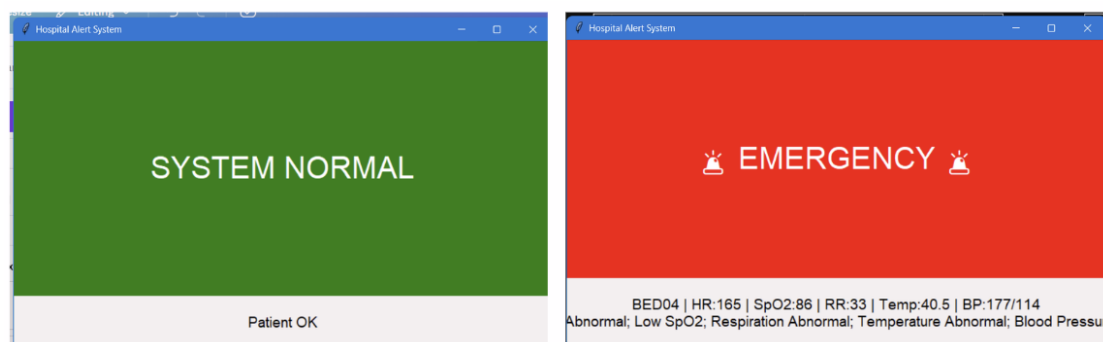


Figure 4.5: System Output: a. during Normal State b. during Alert State

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

In this project, a data logging system for monitoring physiological parameters was developed using the TM4C123GXL microcontroller platform. The primary objective of the system was to simulate physiological data and process it within a monitoring framework capable of visualizing and analyzing the generated parameters.

During Phase 1 of the project, physiological parameters such as Heart Rate, SpO₂, and Body Temperature were simulated using the microcontroller. The generated data was then transmitted to a monitoring system where it was processed and visualized using graphical plots. This enabled real-time observation of parameter variations and helped in identifying abnormal conditions effectively.

An alert mechanism was also implemented to enhance system functionality. Whenever abnormal values were detected, alert messages were generated and broadcast to all systems running the receiver program within the local network. This feature demonstrates the capability of the system to support real-time monitoring and rapid notification.

Overall, the project successfully demonstrates the feasibility of building a physiological data logging and monitoring framework using embedded systems and software-based visualization tools. The developed system provides a foundational platform for further development of real-time healthcare monitoring solutions.

5.2 Future Scope

Although the current implementation demonstrates the basic functionality of physiological data logging and monitoring, several improvements and extensions can be made in future work.

In the next phase of the project, real biomedical sensors can be integrated with the microcontroller to acquire actual physiological signals instead of simulated data. This would allow the system to operate as a complete patient monitoring solution.

The system can also be extended by including additional physiological parameters such as respiratory rate, blood pressure, and electrocardiogram (ECG) signals. Incorporating these parameters would improve the comprehensiveness of the monitoring system.

Another potential enhancement is the integration of secure cloud-based storage and remote access capabilities, enabling healthcare professionals to monitor patient data from remote locations. Furthermore, advanced data analysis techniques such as machine learning can be applied to detect anomalies and predict potential health risks.

These improvements would significantly enhance the capabilities of the proposed system and contribute toward the development of efficient and scalable healthcare monitoring solutions.

References

- C. K. Anju and S. Lekshmi. Wireless biomedical parameter monitoring system using arm microcontroller. *IOSR Journal of Engineering*, 4(5), 2014. URL [https://iosrjen.org/Papers/vol4_issue5%20\(part-3\)/A04530105.pdf](https://iosrjen.org/Papers/vol4_issue5%20(part-3)/A04530105.pdf).
- Chokri Baccouch and Chaima Bahar. Embedded system for ecg monitoring and fatigue detection using machine learning. *IJACSA*, 16(9), 2025. URL https://thesai.org/Downloads/Volume16No9/Paper_7-Embedded_System_for_ECG_Signal_Monitoring.pdf.
- Paolo Bonato. Wearable sensors and systems. *IEEE Engineering in Medicine and Biology Magazine*, 2010. URL <https://dspace.mit.edu/bitstream/handle/1721.1/69890/Bonato-2010-Wearable%20sensors%20and%20systems.pdf>.
- Chao Chen and Carlos Pomalaza-Raez. Implementing and evaluating a wireless body sensor system for automated physiological data acquisition at home. *arXiv preprint arXiv:1006.1178*, 2010. URL <https://arxiv.org/abs/1006.1178>.
- Robert Gao. A microcontroller-based data logger for physiological sensing. *ResearchGate*, 2006. URL <https://www.researchgate.net/publication/3950584>.
- Josiah Hester et al. Self-powered analog sensor data logging device. *arXiv preprint arXiv:2004.12256*, 2020. URL <https://arxiv.org/pdf/2004.12256.pdf>.
- L. Malathi, R. Pavithra, M. Ramya, S. Sandhiya, and V. Varsha. Embedded smart healthcare monitoring using iot (arduino). *IJISSET*, 8, 2021. URL https://ijiset.com/vol18/v8s3/IJISSET_V8_I03_41.pdf.
- Hovannes Mamaghanian et al. Compressed sensing for energy efficient telemonitoring of physiological signals. *arXiv preprint arXiv:1309.7843*, 2013. URL <https://arxiv.org/pdf/1309.7843.pdf>.

- Chris Otto, Aleksandar Milenkovic, Corey Sanders, and Emil Jovanov. System architecture of a wireless body area sensor network for ubiquitous health monitoring. *Journal of Mobile Multimedia*, 2006. URL https://www.researchgate.net/publication/228878299_System_Architecture_of_a_Wireless_Body_Area_Sensor_Network_for_Ubiquitous_Health_Monitoring.
- Alexandros Pantelopoulos and Nikolaos Bourbakis. A survey on wearable sensor-based systems for health monitoring. *IEEE Transactions on Systems, Man, and Cybernetics*, 2010. URL <https://ieeexplore.ieee.org/document/5290333>.
- Amir Rahmani et al. Ubiquitous healthcare monitoring system using integrated sensors. *arXiv preprint arXiv:1309.1542*, 2013. URL <https://arxiv.org/pdf/1309.1542.pdf>.
- Tanmay Sarkar et al. Low cost pc based real time data logging system for slowly varying signals. *arXiv preprint arXiv:1207.2739*, 2012. URL <https://arxiv.org/pdf/1207.2739.pdf>.
- Dimitrios Vergados et al. Ubiquitous mobile health monitoring system for elderly. *arXiv preprint arXiv:1107.3695*, 2011. URL <https://arxiv.org/pdf/1107.3695.pdf>.
- Guang-Zhong Yang et al. Real-time wearable physiological monitoring system for home healthcare. *Sensors*, 22(1), 2022. URL <https://www.mdpi.com/1424-8220/22/1/104/pdf>.

Acknowledgement

We would like to express our deepest appreciation to all those who provided us the opportunity to complete this project. We extend our sincere gratitude to our project guide, **Prof. Sudhakar Sir**, for his invaluable guidance, technical insights, and constant encouragement throughout the design and development of the project **Data Logging System for Physiological Parameters**.

We are also deeply thankful to our Class Advisor, **Prof. Samidha Vengurlekar**, for her constant support, coordination, and motivation throughout the academic session.

We also wish to express our thanks to the Head of the Computer Engineering Department, **Dr. Rachana Dhannawat**, and the Principal of **Usha Mittal Institute of Technology**, **Dr. Yogesh Nerkar**, for providing the necessary resources and an academic environment conducive to project-based learning.

We are grateful to our peers and colleagues for their constructive feedback and support. Finally, we would like to thank our families for their patience and moral support, which were instrumental in the successful completion of this project.

Date: 13 March 2026

Signature: _____
Divya Gurav

Signature: _____
Maithilee Kubal

Signature: _____
Anwasha Jana